# Stanley Chueh 闕楷宸

**Robot Learning · Vision-Language-Action Models · Sim-to-Real Manipulation**

t114318047@ntut.org.tw · +886 978 455 076 · Taipei, Taiwan · stanleychueh.com
github.com/StanleyChueh · linkedin.com/in/stanley-chueh · youtube.com/@Stanley_Chueh

## EDUCATION

**M.S. Electrical Engineering**, National Taipei University of Technology — Sep 2025 – Present
**GPA 4.0 · Rank 2 / 131 (top 1.5%)** · *Research: VLA models, mechanistic interpretability of robot policies*

**B.S. Electrical Engineering**, National Taipei University of Technology — Sep 2021 – Jun 2025
**GPA 3.72** · *Focus: computer vision, robot navigation*

## EXPERIENCE

**Research Assistant**, City Science Lab @ Taipei Tech — Sep 2023 – Present

- Co-lead the lab's Vision-Language-Action (VLA) project; own the simulation, teleoperation, and real-robot deployment stack for the OpenArm bimanual platform.
- Built robot-learning pipelines across three arms (OpenArm, Franka Panda, Koch) and navigation systems for a quadruped (visual SLAM) and a TurtleBot3 (ROS 2 multi-map switching on Nav2).

**Research Assistant**, InstAI (remote) — Aug 2023 – Nov 2024

- Built an edge-AI image recognition pipeline: data capture, augmentation, training, and on-device deployment.

## SELECTED PROJECTS

**Sim-to-Real VLA for Bimanual Manipulation** · OpenArm · project · video — 2026

- Trained a SmolVLA policy for a pick-and-handover task using **only simulated data**: VR teleoperation in Isaac Sim, then Isaac Lab Mimic to expand **10 → ~400 demos**.
- Reached **80% success in simulation** (100 rollouts) and **40% on the real robot** (10 rollouts) at ~30 Hz.

**Language-Conditioned Multi-Task VLA** · Koch arm · project · video — 2026

- Fine-tuned SmolVLA on 60–100 teleoperated demos per task; the robot switches tasks in real time from natural-language commands at ~30 Hz.
- Built real-time cross-attention visualization to inspect what the policy attends to during execution.

**Imitation Learning on Franka Emika Panda** · project · video — 2025

- Set up GELLO leader-follower teleoperation with joint impedance control (Franka ROS); trained ACT on ~100 demos/task and deployed drawer-opening and pick-and-place on the real robot.

**Visual Navigation for a Quadruped** · Triceratops · project · video — 2024

- Designed the navigation stack (Isaac ROS Visual SLAM, AprilTag-assisted localization, depth-to-laserscan local planner) and deployed it on the real robot indoors.

## SKILLS

| | |
|---|---|
| **Robot learning** | LeRobot, ACT, SmolVLA, imitation learning, Isaac Lab Mimic, PyTorch |
| **Robotics** | ROS / ROS 2, Nav2, Franka ROS, Visual SLAM, AprilTag, teleoperation (GELLO, VR) |
| **Simulation** | Isaac Sim, Isaac Lab, Gazebo |
| **Programming** | Python, C / C++ |
| **Languages** | Mandarin (native), English (fluent) |

## AWARDS & ACTIVITIES

**Merit Award**, IWCE Paper Competition — Dec 2023
**Participant**, NTUT PBL International Workshop (3 years) — 2023 – 2025